HIGH-ALTITUDE OCULOMICS: AUTOMATED AI-EXTRACTED RETINAL VESSEL METRICS FROM FUNDUS IMAGES COLLECTED ON MOUNT KILIMANJARO

2257-0689

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Purpose

- To characterize retinal vascular responses to altitude hypoxia using Artificial Intelligence (AI)-derived metrics
- The paradigm supports the use of oculomics to monitor acute physiologic states such as hypoxia

Methods

- Observational field study during a 7-day expedition on Mount Kilimanjaro.
- Twenty climbers (aged 16–68 years) were imaged using a handheld 45° fundus camera (Volk Viva) at baseline (1,340 m) and multiple altitude points during ascent.
- Two AI pipelines, AutoMorph, VascX, extracted vascular metrics from longitudinally registered images (using EyeLiner).

Evaluation

- Mixed-effects models tests day and linear altitude associations.
- Benjamini-Hochberg was used for false discovery rate (FDR) correction.

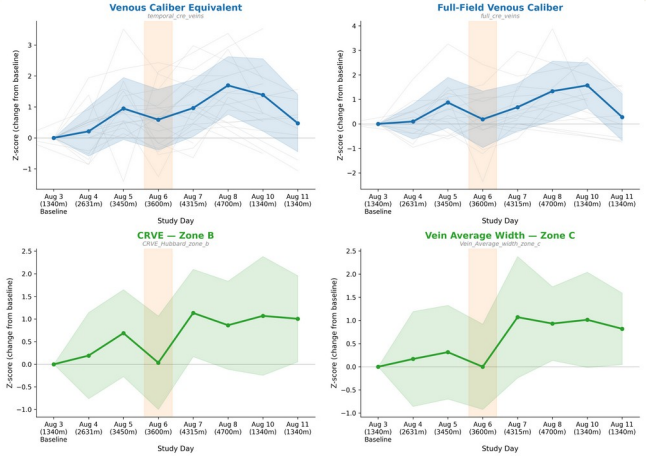

Figure 1. Selection of VascX (blue) and AutoMorph (green) metrics

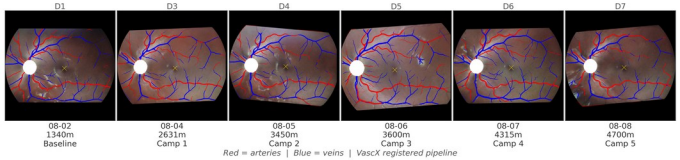

Figure 2. Segmentation and feature extraction from VascX

Results

- Venous caliber and diameter increased consistently with altitude across both pipelines, while arterial changes were less robust (FDR $q<0.05$).
- A non-monotonic dip across multiple metric domains was observed at Camp 3 (3600 m), with recovery at higher altitudes in the categorical model.
- Responses peaked at 4700 m and remained elevated after descent.

Conclusions

- AI-derived retinal metrics reveal a reproducible, vein-dominant hypoxic response with non-monotonic dynamics and post-descent persistence.
- These findings support their potential as noninvasive biomarkers of hypoxic vascular adaptation

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Mixed-effects models tested day and linear altitude associations.
Benjamini-Hochberg FDR.